BITS

BITS CRMPBX

Product Manual

Customer service CRM with Asterisk PBX, call recordings and GoIP GSM gateway support.

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BITS CRMPBX — Product Manual

Customer service CRM with Asterisk PBX, call recordings and GoIP GSM gateway support.

This is the manual for the people who *use* the product. It is not the architecture document, not the deployment guide and not the runbook — those exist and are listed at the end for the people who need them.

What the product is

A service desk for an internet service provider. It brings three things into one console:

- the **subscriber directory** — the customers, their services and their history;
- the **ticket workflow** — the problems, who owns them and how they were resolved;
- the **telephony** — the Asterisk PBX: the softphone, the call log, the recordings and the live monitor.

The desk can also run a **chat** channel on the client's website, but that is an optional add-on which is switched on per client by the vendor.

Who this manual is for

The product has **two access layers**, and a person only ever sees one of them.

	The client tier	The vendor tier
Who	the ISP's own staff	BITS, the vendor of the software
Signs in at	`/login` with a CRM account	`/login` with a vendor account
Sees	its own customers, tickets, calls and agents	every client account, licences, the feature set and the support queue
Documented in	the agent, the desk, the administration	the vendor tier

A client administrator has no vendor permission at all: the vendor console answers 403 and never appears in the navigation. The two layers never share a screen.

The roles inside the client tier

Role	What they are for
Client Admin	Everything inside the organisation: users, roles, settings, branding, the Asterisk endpoint, the GoIP gateways, the recordings
Manager	The service desk: tickets, agents, the recordings, the reports, the live monitor
Supervisor	A team lead: the team's tickets and calls, the live monitor, the reports
QA	Reviews calls and tickets and signs off the checks. QA may listen to the recordings of the calls under review
Agent	The customer on the phone: their tickets, their calls, the customer directory

A person may hold more than one role. A **supervisor** who also needs to listen to a call recording is given the *Recordings* permission by the administrator — see the administration.

Signing in

The console is at the address your administrator gave you, for example `https://crm.example.com/login`.

1. Enter your **username** and **password** and press **Sign in**.
2. A wrong password, or too many wrong passwords, locks the account for a few minutes. This is deliberate: it stops a password being guessed.
3. Your account also carries your **phone extension**. An administrator assigns it, and it is the number your calls go out from.

The **softphone** is a separate sign-in: extension and a PIN, with no CRM account at all. See the agent → the standalone softphone.

The navigation

Everything is in the left-hand sidebar. What you see depends on your role.

Section	Item	Who
Service desk	Dashboard	everyone
	Customers	everyone
	Tickets	everyone
Telephony	Call Logs	everyone
	Live Monitor	supervisor, manager, admin
	Recordings	manager, admin, QA — and only while the switch is on
Infrastructure	Agents	supervisor, manager, admin
	Asterisk PBX	supervisor, manager, admin
	GoIP Gateways	supervisor, manager, admin
Administration	Reports	supervisor, manager, admin
	Audit Trail	supervisor, manager, admin
	Users & Roles	admin
	Settings	admin

Two more places are reachable but not in the sidebar: **Profile** (your own details, your permissions and your recent activity) and **About this app** (the product name, the company and the copyright).

The manual

Part	For
01 — The agent	The customer on the phone: the console, the customer lookup, taking a call, tickets, the softphone, the chat
02 — The desk	The supervisor, the manager and QA: the dashboard, the live monitor, the recordings, the reports, the audit trail
03 — The administration	The client administrator: users and roles, the settings, the branding, the PBX, the GoIP gateways, the customer import
04 — The vendor tier	BITS: the client accounts, the licences, the feature set, the optional add-ons, the support queue
05 — Reference	The glossary, the troubleshooting tables and the contacts

The documentation for the people who run it

These are not the manual. They are for the operator, the developer and the person who has to fix something at half past midnight.

Document	For
README	What the repository is, how to run it locally, the test suite
Deployment	Putting the product into service
Environment	Every setting and every `.env` variable
Operations	Running it day to day: the worker, the schedules, the health
Runbook	What to do when something breaks
Architecture	How it is built, and why
ADR 0001–0016	The decisions, and the reasons for them

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01 — The agent

The agent is the customer on the phone. This part covers the working day: finding the customer, taking the call, logging it, and the tickets that come out of it.

The console

Call Logs → **Softphone** is the call console. It is the screen you live in while the phone is ringing. Three zones, top to bottom:

1. **The customer context** — a search box at the top. Type a name, an account number or a phone number and press **Look up**. Clicking a result pins the customer to the console: their name, their account number and their number appear above the call controls for the rest of the call.
2. **Call control** — the blue bar. The destination number, the **Ext** box for a transfer, **Transfer**, **Hang up** and the timer.
3. **Disposition** — what happened on the call, and the notes you wrote. This is the part that is saved.

The keypad

The destination number has a **keypad** of its own, beside it. Press the keyboard button and the pad appears; press it again and the pad goes away. The pad types into the number field, so the field, the customer lookup and the call always agree on what is being dialled.

There is also a backspace button at the right of the number field, so a mistyped digit is one click to remove.

> The pad is shown by default. If you hide it, the choice is remembered for your browser.

Placing a call

1. Look the customer up, or type the number in the destination field.
2. Press **Call**. The bar turns from blue to green and the timer starts.
3. While the call runs you have:

Control	What it does
Hang up	Ends the call
Transfer	Asks for an extension and hands the call over
Ext	The extension to transfer to, filled in before you press Transfer

4. When the call ends, pick an **Outcome** and write **Notes**.
5. Press **Save disposition**.

The outcomes are set by your administrator. The usual ones are:

Outcome	When
Resolved	The call answered the customer's problem
Follow-up required	Something came out of it that needs a ticket
No answer / Line busy / Uncontactable	Nobody answered
Callback requested	The customer asked you to call back
Escalated to technical	It needs the engineers
Billing / credit request	It is really a billing matter
Technical issue reported	The customer reported a fault
Plan inquiry / upgrade	They asked about their plan
Wrong number / Duplicate record	The number or the record was wrong

The call, its duration and its recording are written against your account. You do not have to do anything else: the ticket, if there is one, is found from the customer.

Customers

Customers is the subscriber directory. It is a long list, so it searches as you type: a name, an account number or a phone number.

A customer page holds four cards:

Card	What is in it
Contact details	Name, business name, type, status, the numbers, the e-mail, the address, the balance, the date they joined
Subscriptions	Their service: the plan, the IP address, the router, the monthly price and the balance
Tickets	The tickets they have raised
Call history	The calls they have made and received, with the disposition of each

From the customer page you can **Call** a customer straight away — the console opens with them already pinned.

The account number and the cell number are the two things worth reading out loud, because they are what the customer gives you when they call back.

Tickets

A ticket is a problem with an owner and a status.

- **Tickets** lists them, with filters for status, priority, category and assignment.
- A **ticket** page holds the description, the comments, the activity trail (who did what and when), the customer, and the properties: **status**, **assign to** and the priority.

The workflow:

Status	Means
Open	Nobody has picked it up yet
In progress	Somebody is on it
Waiting on customer	It is with the customer; the clock is on them
Escalated	It needs a manager
Resolved	It is done — the customer is waiting for the confirmation
Closed	It is finished and the customer has answered or the period has run out

Assign to puts your name on it. An agent can work their own tickets; a supervisor can assign to anyone.

The standalone softphone

Some desks want a telephone of their own, away from the console: a tablet on the desk with just the phone on it, no customers and no tickets in sight.

`/softphone` is that screen. It signs in with your **extension and a PIN** — no CRM account, no customer data. It places calls from your extension and shows your own calls. That is all it does.

Installing it

The softphone is a program, not a web page. On the sign-in screen there is an **Install it as an app** card:

- press **Install the softphone**, and the browser does the rest;
- or use the browser's own menu — in Edge, the **Apps** button in the toolbar (the squares), then **Install this site as an app**; in Chrome, the install icon at the right of the address bar.

Installed, it opens in a window of its own with its own icon on the desktop and in the task bar: no address bar, no tabs, no bookmarks. The install offer disappears once it is installed.

The size

A telephone is a small thing, so the softphone is drawn at its own width whatever size the window is:

Size	Width	For
Compact	320 px	Fitting it beside something else on the screen
Normal	380 px	The default — the size of a desk phone
Large	460 px	More room for the number
Full screen	The whole window	With the call list beside the dialler

The **compress button** in the title bar cycles the size, and **Settings** → **Size** has the full list. The choice is remembered for your extension, so it follows you to the next machine.

The themes

Five themes, and each is picked for the room it is used in:

Theme	For
Midnight	The default — deep indigo, low light, long shifts
Graphite	Business, daylight, no colour noise
Aurora	Teal and violet; calm, a little unusual
Ember	Warm amber on charcoal, for the late shift
Daylight	Bright, for an open office with the lights on

The theme is chosen from the softphone itself (**Settings** → **Theme**) and is remembered for your extension. A deployment sets the default for everyone.

The controls

While a call runs, the dialler is replaced by the call screen: the number, the timer, the recording badge and the round controls.

Control	What it does
Mute	Turns your microphone off. The caller hears silence; the call stays up
Hold	Puts the call on hold and plays the hold music. The caller hears music; you can take another call
Transfer	Hands the call to an extension
Speaker	Plays the call on the computer's loudspeaker — for a handset-free desk
Hang up	Ends the call

Hold and mute show what the **PBX** did, not what was asked for. If the switchboard refuses an operation, the button says so rather than pretending it worked.

The **speaker** button only appears on a machine that actually has an audio output. If the computer has no speakers at all, the button is not offered.

> The softphone has **no recordings**. A recording is an account matter: the manager and QA listen to them from the console. Your calls are still recorded and attached to the call, but you cannot play them back from the softphone — and that is on purpose.

The chat channel (an optional add-on)

If the client has bought the chat channel, a chat button appears on their website. A visitor presses it, types a message, and the conversation appears in **Chats** in the console.

- **Chats** lists the conversations; the busiest ones first.
- A **conversation** holds the visitor's details, the whole message history, and a box to answer in.
- **Assigned to** puts an agent's name on it.

The visitor is matched to a customer by their e-mail address, so the conversation lands on the right record without the visitor having to know anything about it.

If the client has not bought the chat channel, the button is not there, the widget answers 404, and the console answers 403. Nothing half-works.

What the agent cannot do

This is deliberate, and it is there so that a mistake at a desk cannot become a breach:

- **The agent cannot open the recordings** — not from the console, not from the softphone. The manager and QA can.
- **The agent cannot see the audit trail**, the reports, the other agents or the settings.
- **The agent cannot change a customer's record** beyond what their permission allows, and every change is written to the audit trail with your name against it.

If you need something from this list, ask your supervisor or the administrator.

02 — The desk

The supervisor, the manager and QA. This part covers what the people who run the desk see: the morning's numbers, the live monitor, the recordings, the reports and the audit trail.

The dashboard

Dashboard is the first screen after signing in. It answers three questions: what happened lately, what needs attention now, and who is on the desk.

Card	What it shows
Recent calls	The last calls across the desk, with their number, their duration and their disposition
Tickets needing attention	The open and escalated tickets, oldest and most urgent first
Agent presence	Who is on the desk right now, and in which state
Ticket status breakdown	How the open tickets are spread over the statuses
Top categories	The categories the desk is spending its time on

The ticket status breakdown and the top categories are the two worth looking at every morning: they say whether the desk is being pulled towards one kind of problem.

Live Monitor

Live Monitor is the wall of the call centre: every channel currently on the PBX, who is on it, and whether it is being recorded.

- A **channel** row shows the channel, the agent on it, the number dialled, how long it has run and whether it is **REC** — being recorded.
- The **agent presence** panel shows every extension and its state, so you can see who is free, who is on a call and who is logged out.

This screen is for the supervisor who has to know, *now*, where the desk is. It refreshes every few seconds.

> The monitor shows that a call **is** being recorded. It is not a way to listen to it — that is the Recordings screen, and it has its own rules (below).

Call Logs

Call Logs is the history: every call, with the number, the direction, the duration, the disposition and the agent. The filters narrow it by number, agent, status and date.

A **call** page holds the call detail, the recording (if the agent may listen), the disposition and the links to the customer and the ticket.

Recordings

Recordings is the library of the recorded calls, with the retention sweep at the bottom.

Who may listen

This is deliberately narrow, and it is worth knowing before you go looking for something:

Role	May listen
Agent	**no**
Supervisor	**no**
QA	**yes** — reviewing a call and signing it off means hearing it
Manager	yes
Client Admin	yes
Vendor	yes

And on top of that there is a **switch**: **Settings** → **Recordings** → ***Let the team listen to the call recordings***. While it is off, nobody in the client can open a recording at all — the calls are still recorded and kept, they just cannot be reached. The vendor can always see them.

Both the permission and the switch are checked on the route, so a typed URL is refused exactly like a hidden menu item.

> If a supervisor needs to listen to their team's calls, the administrator gives them the *Recordings* permission in **Users & Roles**. If the desk decides it does not want call audio at all for a while, the switch goes off.

The retention sweep

The sweep deletes every recording older than the number of days given. It is how the library is kept inside the size the client has agreed. The administrator sets the number of days in **Settings** → **Recordings**.

Reports

Reports is the desk's month in one screen:

Report	What it answers
Daily call volume	How many calls came in and went out each day
Agent performance	Who handled what
Ticket status	Where the tickets are sitting
Top categories	What the customers are calling about
Ticket snapshot	The tickets as they stand now

Two exports sit alongside them: **calls.csv** and **tickets.csv**, for spreadsheets and for the client's own reporting.

The audit trail

Audit Trail records every change made in the console: who did it, what they changed, from which address and when. It cannot be edited from the application, and it is where an administrator looks when somebody asks *who changed that customer's number*.

The filters narrow it by the person, the action, the entity and the date.

Agents

Agents is the roster: every agent, their extension, their presence, and the PBX as it sees them.

- **Assign extension** gives an agent their extension and their softphone PIN.
- **Presence** shows who is free, busy, ringing or logged out.
- The **PBX snapshot** shows the extensions the switch actually has, next to the ones the CRM believes in — the two drifting apart is a common cause of "my phone does not ring".

QA

QA is a role, not a screen. In the console it adds two things:

- QA may **listen to the recordings** of the calls under review. The *Recordings* permission is what carries that, and the administrator decides whether the desk's switch is on.
- QA holds the *Sign off QA checks* permission, which is what lets QA close a ticket that has been checked. It is granted to QA, to the manager and to the administrator; it is not granted to the agent or the supervisor.

The ticket statuses are the same as everybody else's — there is no separate "verified" status — so a QA check is signalled by moving the ticket to **Resolved** and writing the outcome in the comment. Every change is written to the audit trail with the name of whoever made it against it.

03 — The administration

This part is for the **client administrator**: the person who owns the installation. They set the users up, they point the product at the PBX, they decide what the brand looks like, and they decide who may listen to the recordings.

Users & Roles

Users & Roles has three tabs.

Accounts

Every account in the organisation:

Column	What it is
The person	Their name and their username
The role	What they may do
The status	Active, or locked out
The last sign-in	When they were last here

Creating an account asks for the name, the username, the password and the roles. The password is never displayed again — it is stored hashed, and the administrator resets it rather than reads it.

Roles

The five built-in roles, with what each one is for:

Role	For
Client Admin	Everything inside the organisation
Manager	The service desk: tickets, agents, recordings, reports, the PBX and the gateways
Supervisor	A team lead
QA	Reviews calls and tickets, signs off checks, reads the recordings
Agent	The customer on the phone

A person may hold more than one role, and that is how a supervisor who also needs to listen to a team's calls gets that access.

The permission catalogue

The third tab lists every permission the product has, grouped, with a tick against the roles that hold it. It is the quickest way to answer "who can do this?" without reading the role list.

> The built-in roles are **authoritative**: the permissions they hold are written from the code every time the migration runs, so the database always matches the product. A custom role is not affected by that.

Settings

Settings is where the installation is configured. Each card is a group of settings, and each card has its own **Save** button.

Branding

Everything the customer sees, with a live preview on the right.

Setting	What it changes
Application name	The product's name in the header, the sign-in screen and the reports
Short name	The name on the task bar and in the installed app
Company name	The legal name, on the footer and on the recordings
Vendor name	The name of the vendor of the software — this is **BITS**, never the ISP
Login subtitle	An optional line under the logo on the sign-in screen
Primary colour, **Accent colour**	The two colours of the console and of the sign-in screen
Support phone, **Support e-mail**	Printed on the customer-facing screens
Footer note	A line in the footer
Website	The vendor's website, shown on the About page
Console theme	The default theme: Ocean, Graphite, Emerald or Daylight

An agent can change the console theme for themselves from the navbar; the setting here is the default everybody starts with.

Asterisk

Where the product finds the PBX.

Setting	What it is
Asterisk enabled	The master switch. Off, the product still runs but cannot place or answer calls
ARI url, **ARI user**, **ARI password**	The REST interface — this is how the product places and controls calls
AMI host, **AMI port**, **AMI user**, **AMI password**	The manager interface — used for presence, and as a fallback when ARI is unavailable
Asterisk app name	The name the event worker registers under
SIP password secret	The secret the SIP passwords are derived from. Left blank, it is derived from the application's own key
Recording auto record	Record every agent call automatically
Recording retention days	How long a recording is kept before the cleanup removes it
Recording silence seconds	The padding at the start and the end of a recording

> Leave the endpoint and the credentials **empty** and the deployment's own `.env` decides them. A row here overrides the `.env`, which is what you want when the administrator changes the box rather than the server.

Recordings

One setting, and it is the important one:

> **Let the team listen to the call recordings**

- **On** — the roles that hold the *Recordings* permission can listen.
- **Off** — nobody in the client can open a recording. The calls are still recorded and kept; they simply cannot be reached.

Both the switch and the permission are checked on the route, so a typed URL is refused exactly like a hidden menu item. See the desk → recordings for who holds the permission.

Softphone

Setting	What it is
Softphone theme	The default theme: Midnight, Graphite, Aurora, Ember or Daylight
Softphone size	The default size: Compact, Normal, Large or Full screen
Softphone session hours	How long a softphone sign-in stays valid
Softphone speaker enabled	Whether the speaker control is offered at all
Moh class	The music-on-hold class used when a call is put on hold

An agent picks their own theme and size on the softphone; these are the defaults.

Tickets, customers and GoIP

- **Tickets** — the first-response and resolution SLA targets for urgent tickets, and the prefix on generated ticket numbers.
- **Customers** — the prefix on generated customer numbers.
- **GoIP** — the master switch for the GSM gateway integration, and the transport it uses.

The Asterisk PBX screen

Asterisk PBX shows the connection and what the switch is doing:

- **Connection** — whether the ARI interface answers, with the Asterisk version and the number of endpoints it reports.
- **Agent presence** — every extension and its state.
- **Live channels** — the calls currently on the switch.
- **Connection settings** — the endpoint and the credentials, on the same screen as the status, so there is one place to look when a call will not go out.

If **Connection** is red, the product cannot reach the PBX: check the URL, the user and the password, and that the ARI user is allowed to connect.

GoIP Gateways

GoIP Gateways is the GSM gateway side: the boxes, their SIM slots, their state and the **SMS inbox**.

A GoIP box is a GSM-to-SIP gateway: each SIM slot registers with Asterisk as an endpoint, so the desk can place and receive calls on a mobile number. Each slot needs its own number, and the SMS inbox is where the messages arrive.

Importing the customers

A service desk needs its subscribers before it can take a call. The legacy CRM's customers are imported from its SQL dump:

```
cd /var/www/html/crm

# 1. Read it first - nothing is written without --apply.
php bin/import-legacy-customers.php

# 2. Write the customers.
php bin/import-legacy-customers.php --apply
```

The importer reads the dump file directly, so there is nothing to restore into a database first.

Option	What it does
(none)	A dry run: prints the field map, the counts and a masked sample. Nothing is written
`--apply`	Writes the customers
`--limit=50`	Only the first 50 rows
`--csv=/tmp/imported.csv`	Also writes a reconciliation list. It holds personal data — keep it out of the repository
(a path)	Read a dump from somewhere else

The import is an **upsert on the account number**, so it can be run twice: a customer that is already in the CRM is updated, never duplicated.

Check the result:

```
SELECT COUNT(*) AS customers, SUM(status = 'active') AS active FROM customers;
```

A note on what is switched off

Three things in this product are switched off or restricted on purpose, and an administrator should know about all three:

	Where	Why
The **chat channel**	The vendor's feature set	It is an optional add-on, switched on per client when the client takes it
The **recordings**	Settings → Recordings	A recording is an account matter; the administration decides whether the team hears them at all
The **vendor console**	—	A client administrator has no vendor permission; the two layers never share a screen

04 — The vendor tier

This part is for **BITS**, the vendor of the software. The vendor tier is a separate access layer: a vendor account sees every client account, the licences, the feature set and the support queue, and it never shares a screen with the client tier.

The vendor dashboard

Vendor is the landing screen. Five things are on it:

Card	What it is for
The licence	The licence this installation runs under, and the button to issue a new one
Feature set	The product features, by module, and the switches that turn them off for a client
Optional add-ons	The features that are not part of every client's licence — the chat channel is the first — and the switches that switch them on
Vendor support queue	The tickets raised against the product
Client accounts	Every client organisation on the platform

Client accounts

Client accounts lists the client organisations: who they are, how many accounts and extensions they have, and when they were added. Opening one shows their licence, their feature set and their support tickets.

A client organisation is a **scope**: its users, its customers, its calls and its recordings belong to it and to nobody else. A vendor sees across the scopes; a client never does.

The licence

A licence says what an installation is allowed to do:

Field	What it decides
The key	The installation's identity
The plan	Which features are available
The seats	How many people may sign in
The expiry	When it stops

The licence is displayed on the vendor dashboard and on the client administration's own screen. **A licence does not lock the product**: it records what has been agreed, and the feature set is what actually decides what a client can reach. That is deliberate — an installation must never become unusable because a licence row was edited.

The feature set

Features is the product, broken into modules — CRM, telephony, gateway, core — with a switch against each one.

- **On** — the client sees the screen and the feature answers its API calls.
- **Off** — the screen is hidden and the feature refuses its API calls, so a rollout can be staged and nothing half-works.
- **Core** — the platform cannot run without it, so the switch is locked on.

The optional add-ons

The add-ons are the features that are **not part of every client's licence**. They have their own card, separate from the product features, because the conversation with a client is: "*do you want chat?*" — and the answer has to be one click on the right row.

Add-on	What it is
Chat engagement	A chat button on the client's website, with the conversations landing in the client's console

An add-on:

- **ships switched off**, so a client never sees a feature they did not buy;
- is **never core**, so the platform cannot depend on it;
- is switched on **for one client** when that client takes it.

When the add-on is off, the visitor's widget answers 404, the console answers 403, and the client sees nothing at all.

Adding a feature

The bottom of the screen adds a feature: its key, its name, what it does, which module it belongs to, whether it is an optional add-on, and whether it starts switched on.

An optional add-on **always** starts switched off, whatever the form says — the vendor then switches it on for the clients that take it. That rule is in the code, not in the form, so it cannot be forgotten.

The support queue

Vendor support tickets are the tickets raised against the product itself, as opposed to the tickets a client raises against their own customers.

- The queue lists them, with their state and who they are assigned to.
- A ticket holds the description, the vendor's response and its properties.
- The client raises one from their own console; the vendor answers here.

Everything the vendor does is written to the audit trail with the vendor account's name against it, exactly as the client tier's actions are.

What the vendor tier is not

- It is **not** a way to read a client's customer data casually. The vendor sees the client accounts, their licences and their feature sets; a client administrator has no vendor permission at all, and the vendor console answers 403 for them.
- It is **not** a back door into the client's recordings. The recordings belong to the client tier and follow the client's own switch.
- It is **not** the place where the product is configured. The client administrator owns the branding, the users, the PBX and the gateways; the vendor decides which features the client may reach.

05 — Reference

The glossary, the troubleshooting tables and the contacts.

Glossary

Term	What it means
ARI	The Asterisk REST Interface — how the product places and controls calls
AMI	The Asterisk Manager Interface — used for presence, and as a fallback when ARI is unavailable
Audit trail	The record of every change made in the console, with the person, the time and the address
Chat	The optional chat channel on the client's website. An add-on, switched on per client
Client tier	The ISP's own staff and their screens. A person only ever sees this tier or the vendor tier
Core feature	A feature the platform cannot run without. Its switch is locked on
Disposition	The outcome written against a call — *Resolved*, *No answer*, *Escalated to technical* and the rest
Extension	The number an agent's calls go out from. One per agent
Feature flag	The row behind a switch in the feature set
GoIP	A GSM-to-SIP gateway box. Each SIM slot registers with Asterisk as an endpoint, so the desk can use a mobile number
Hold	Putting a call on hold. The caller hears the hold music
Moh	Music on hold — the music a held caller hears
Optional add-on	A feature that is not part of every client's licence. It ships switched off and the vendor switches it on per client
PBX	The private branch exchange — the Asterisk box that carries the calls
Recording	The audio of a call, written by the PBX and kept for a number of days
Scope	The boundary a client organisation lives in. Its users, customers and calls belong to it and to nobody else
SIP	The protocol the extensions and the gateways speak
Softphone	The telephone, on its own screen. Signs in with an extension and a PIN, and keeps no recordings
Speaker	The softphone control that plays the live call on the computer's loudspeaker
Transfer	Handing a call to another extension
Vendor tier	BITS's own screens: the client accounts, the licences, the feature set and the support queue

Troubleshooting — the agent

What you see	What it usually means	What to do
The console is blank after signing in	The account has no role	Ask the administrator
Call does nothing	The Asterisk integration is switched off	Ask the administrator; check **Asterisk PBX → Connection**
The call goes out but the customer hears nothing	The microphone is muted, or the agent is on hold	Check the mute and hold buttons on the softphone
The customer hears music	The call is on hold	Take it off hold
The softphone will not sign in	The PIN is wrong, or the extension has no PIN	An administrator sets it on the agent's record
No speaker button on the softphone	The computer has no audio output	Use a handset
The chat button is not on the website	The chat channel is an add-on and the client has not taken it	Ask the vendor
A recording cannot be opened	Recordings are an account matter	Ask the manager or QA

Troubleshooting — the administrator

What you see	What it usually means	What to do
Connection on the Asterisk screen is red	The product cannot reach the PBX	Check the ARI url, the user and the password on **Settings → Asterisk**; check that the ARI user is allowed to connect
The calls work but the presence does not	The AMI details are wrong	Check **Settings → Asterisk** — the AMI host, port, user and password
The extension is in the CRM but the phone does not ring	The PBX and the CRM have drifted apart	Compare **Agents** with **Asterisk PBX → Live channels**; the PBX is the authority
The softphone will not install	The browser did not offer the prompt	Use the browser's own menu: Edge **Apps → Install this site as an app**; Chrome's install icon in the address bar
The softphone looks stretched	The size is set to *Full screen*	The compress button in the title bar, or **Settings → Size**
Nobody can open a recording	The switch is off, or the role has no permission	**Settings → Recordings** for the switch; **Users & Roles** for the permission
A setting will not save	—	Save the card the setting is on; each group of settings has its own **Save** button
The chat is not in the console	The add-on is switched off for this client	The vendor switches it on, per client, in the vendor tier

Who to ask

For	Ask
A customer account, a user's password, a role	The client administrator
The PBX, the extensions, the GoIP boxes	The client administrator, or whoever looks after the Asterisk box
Whether the chat channel can be turned on	**BITS** — it is an add-on, and the vendor decides it per client
A fault in the product itself	Raise a vendor support ticket from the console
A recording	The manager or QA, while the switch is on

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